

A working roadworks planning system

Built and running. Real Melbourne data, simulation and AI driver agents.

REAL DATA FOUNDATION OSM roads | SCATS traffic counts | PTV timetables | City pedestrian counts

OPEN LIVE DEMO ↗

PROBLEM STATEMENT

Closures disrupt **drivers, public transport and pedestrians**. Planners need to test the impact, including **overlapping works**, before deployment.

OUR INNOVATION

Four AI driver perspectives

Commuter / Local / Visitor / Delivery

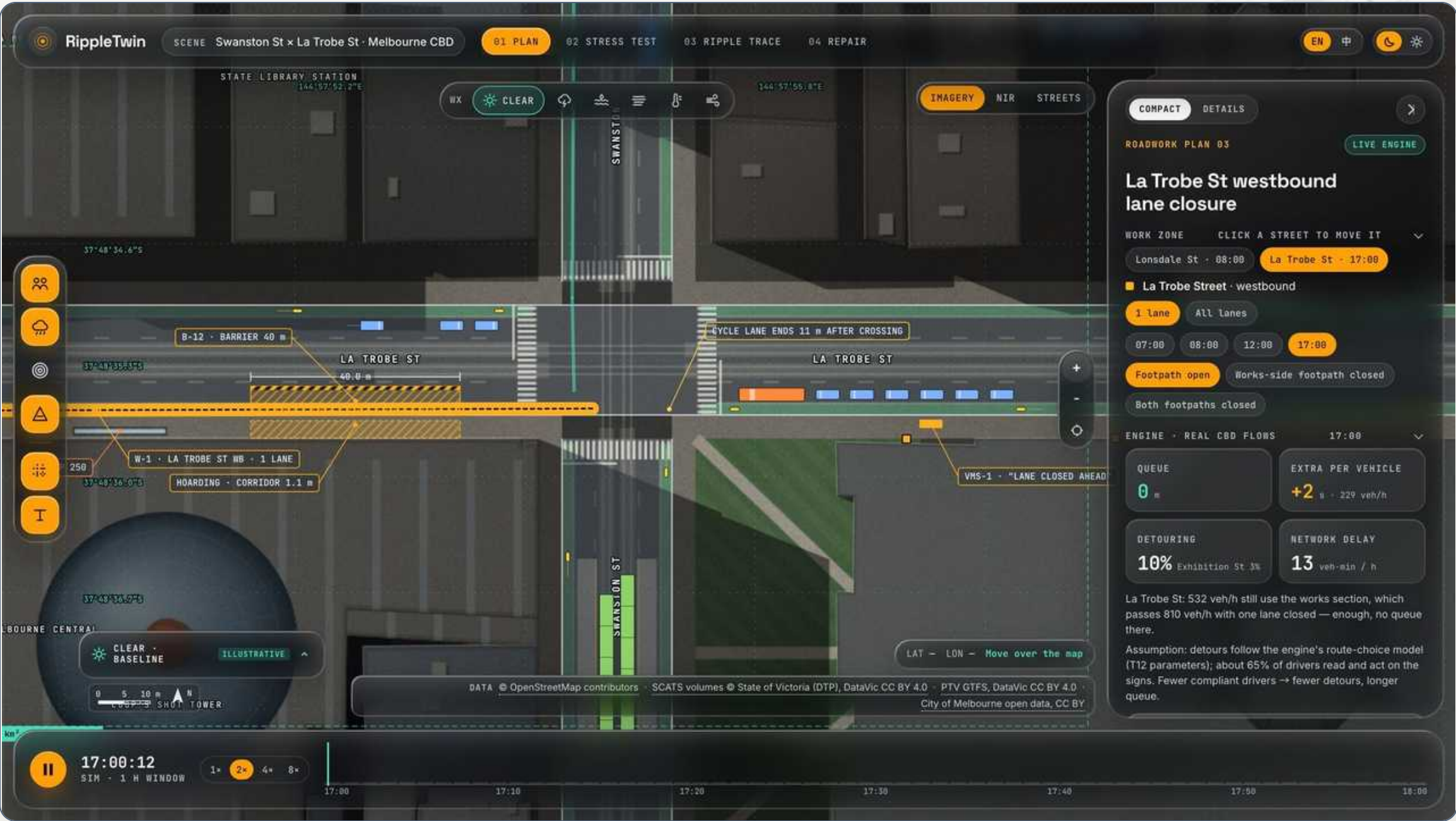
Four LLM driver agents interpret the same sign differently. Their readings inform the simulation.

AI interprets. The engine calculates.

VALUE BEFORE DEPLOYMENT

Helps **councils and contractors** compare **traffic impact and equipment cost** before crews arrive on site.

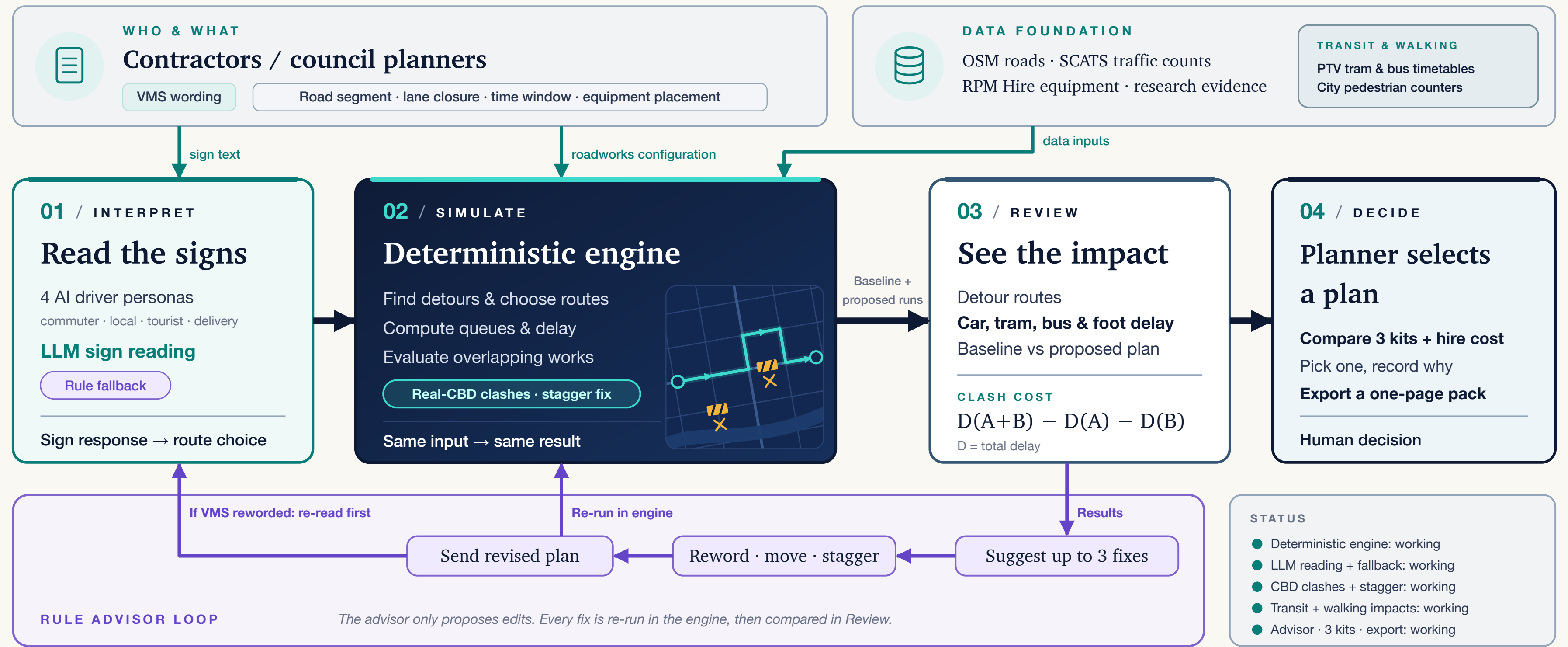
Test a fix. Select a plan. Export an execution pack.



Running junction simulation: vehicles, tram and roadworks on a real map. Movements are illustrative.

System architecture

AI interprets. The engine calculates.



TECH STACK Python / OSMnx data pipeline, JavaScript simulation and Canvas UI, DeepSeek sign reading, Cloudflare Workers.

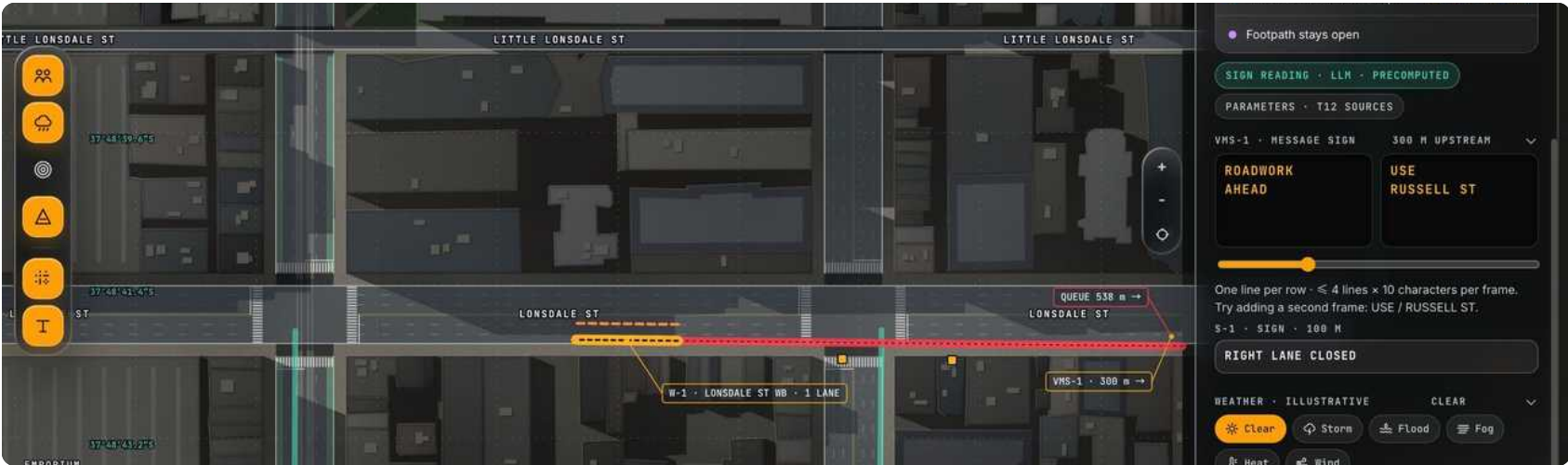
DAY 3 MVP An end-to-end planning demo with scenario comparison and export. Remaining: final validation, demo rehearsal and release packaging.

A working system, end to end

Real product screens. From a road closure to a plan you can act on.

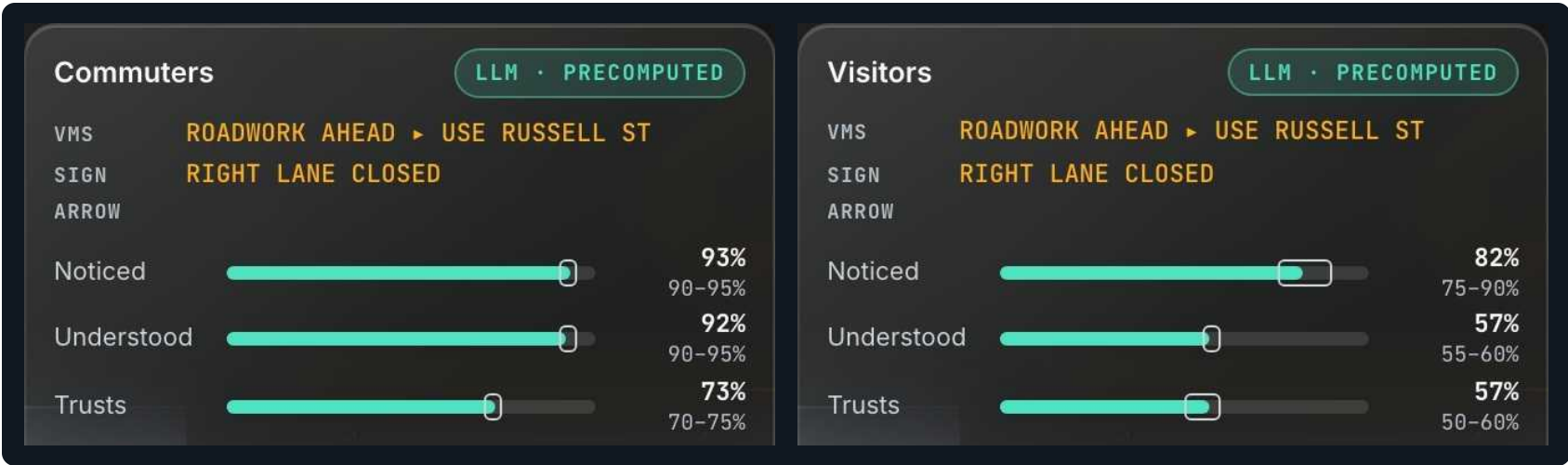
Uncapped / RippleTwin
[Open live demo](#)

01 Configure the site



Set the road, closure and hour. **Test your sign wording.**

02 Understand different drivers



Same sign, different responses. Commuter and visitor AI readings.

03 See the network impact



Trace **queues, detours and clashes** across real Melbourne roads.

04 Compare and export

PLANS COMPARED				
PLAN	CAR DELAY veh·min	TRAM & BUS rider·min	ON FOOT ped·min	HIRE works period
A · Minimum	10,493	18,693	0	A\$515
B · Standard	10,493	18,693	0	A\$1,765
C · Guided CHOSEN	937	1,651	0	A\$1,765

Traffic numbers: simulation engine, weekday 08:00, one hour. Hire: whole works period.

✓ 3 equipment options

✓ Recorded choice

✓ Execution pack

Choose a plan and export it. Impact and hire cost in one view.

-46% vehicle delay with clearer signs¹

3-day shift removes sampled clash cost²

+176 m walking detour made visible¹

MODELLED
SCENARIO RESULTS

¹ Lonsdale St westbound, weekday 08:00, one lane closed. ² Demo works register, 3 overlapping days × 2 peak hours: 27,180 to 0 veh·min. AI cards show precomputed LLM readings, not measured behaviour. Flows include estimates; passenger loads, stock and hire rates include assumptions. PRODUCT / 03